DUY KHUONG

kd23132@engineering.upenn.edu | Portfolio: https://kd-engineering-portfolio.vercel.app/ | 215-868-8693 | Philadelphia, PA 19104

EDUCATION

University of Pennsylvania *Aug 2025 – May 2027*

M.S., Mechanical Engineering and Applied Mechanics

- Concentration: Mechatronic and Robotic Systems

VinUniversity *Sep 2022 – May 2026*

B.S., Mechanical Engineering

- Dean's List, Fall 2023 and Fall 2024; 90% Merit Scholarship for Undergraduate Study

Coursework: Design of Mechatronic Systems, Advanced Mechatronics, Advanced Robotics, Intelligent Physical Systems

SKILLS

- **Software Tools**: Python, R, MATLAB, Git, Bash, Autodesk Fusion, SolidWorks, Ansys FEA, G-Code
- **Hardware Tools**: MCU, IMU, Time of Flight & Ultrasonic Sensor, Vive, CNC Machining, 3D prototyping, DFM/DFA
- **Frameworks & Libraries**: Pytorch, TensorFlow, Numpy, Pandas, FedML, Librosa, MNE, OpenCV, ROS

EXPERIENCE

Security and Artificial Intelligence Lab | VinUniversity *Aug 2024 – May 2025*

Undergraduate Research Assistant

- First author of "*FedKoE: Enhancing Federated Multimodal Learning through Knowledge of Experts*," published in the *FL-AsiaCCS 2025* workshop.
- Developed a federated learning method to quantify each modality's contribution and rebalance training for multimodal classification, outperforming popular methods.
- Evaluated a private multimodal dataset and delivered comprehensive quality/labeling feedback.

PROJECTS

Project Robot Battle Arena | University of Pennsylvania Portfolio/battle_bot

- Designed and assembled a mobile battle robot supporting Bluetooth-controlled mode for manual attacks/movement, and autonomous mode for Vive-based localization and wall-following navigation.
- Engineered a dual-microcontroller system linked via ESP-NOW and I2C for sensor integration. Fused Vive tracking and time-of-flight data with mecanum-wheel kinematics for real-time motor control.

Project Quadcopter Design | VinUniversity Portfolio/quadcopter

- Built a quadcopter using aerodynamic design and control theory to achieve stable hover and waypoint navigation.
- Designed and fabricated a carbon-fiber frame through running ANSYS structural simulations. Iterating 3D-printed prototypes, optimized geometry and CNC-machined carbon-fiber parts to achieve top stability and highest score.

Project Arduino Based: Self Balancing Robot | VinUniversity Portfolio/balance_bot

- Built a two-wheeled self-balancing robot using IMU sensor integration and control design. Enabled autonomous balance and position navigation.
- Led hardware assembly of Arduino Nano MCU, L298N driver, and brushed DC motors. Designed the 3D-printed frame and tuned the PID control parameters. Earned top performance a perfect score (100/100)

Project Barrier Gate | VinUniversity Portfolio/barrier_gate

- Designed and fabricated a motor-driven swing barrier gate using quick-return mechanism. Performed kinematic analysis with MATLAB, CAD modeling, and manufactured components via laser cutting and CNC machining to achieve optimized opening/closing time ratio.

Project Digital Stethoscope | VinUniversity Portfolio/digital_scope

- Designed and assembled a 4-piece 3D-printed enclosure and frame for a wireless digital stethoscope, providing mechanical support and safe packaging for the embedded system.
- Ensured reliable Bluetooth communication between the ESP32 and mobile application for real-time PCG streaming. Trained a lightweight CNN-based classifier using the PhysioNet 2016 Heart Sound Challenge dataset.